

Lab: Movement and Reflexes -- Your Body in Action

Objective

Learning Goal: Investigate how your body takes action through voluntary movements and involuntary reflexes, and discover how the motor system turns predictions into physical actions.

This is an **investigation lab** designed for middle school students (Grades 6-8).

Materials Needed

- Small rubber ball
 - Ruler or meter stick
 - Tape for marking floor distances
 - Lab journal or notebook
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Part 1: Reflex Testing (10 min)

Reflexes are your body's fastest actions -- they happen before your conscious brain even knows what occurred. Test some reflexes with a partner.

Knee-jerk reflex (if comfortable): Sit with one leg crossed over the other. Your partner gently taps just below your kneecap with the side of their hand. What happens?

Reflex Tested	What happened?	Did you control it?	How fast was it?
Knee-jerk			
Blink (partner claps near your face)			
Pupil (shine phone flashlight briefly, then away)			

Write your response here

Part 2: Motor Skill Challenge (15 min)

Voluntary actions require your brain to send precise commands to your muscles. Let's test motor accuracy and see how practice improves it.

Ball toss accuracy: Stand 2 meters from a target (trash can or marked spot). Throw a ball at the target 10 times. Record hits and misses.

Throw	Hit or Miss?	What did you adjust?	Closer or farther than last throw?
1		(baseline)	
2			
3			
4			
5			
6			
7			
8			
9			
10			

Write your response here

Part 3: Action as Prediction Fulfillment (10 min)

In Active Inference, you do not just react -- you act to **MAKE** your predictions come true. Your brain predicts where the ball should go, and your muscles try to make that prediction happen.

Scenario	What does your body predict?	What action makes that prediction true?
You are thirsty		
You are about to trip on a step		
You see a ball flying toward your head		
You are cold in a classroom		

Write your response here

Part 4: Reflection Table

Question	Your Answer
What is the difference between a reflex and a voluntary action?	
How does practice improve motor accuracy?	
How does the motor system use prediction?	
What is Active Inference's view of action in the body?	
Give an example of acting to fulfill a prediction.	

Write your response here